Amin Shabani

Machine Learning Researcher
RBC Borealis, Vancouver
BC, Canada

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Education

- Ph.D. in **Computing Science** — Sep 2019 - Aug 2025
  Simon Fraser University, BC, Canada
- M.S. in **Electrical and Computer Engineering** — Sep 2017 - Aug 2019
  Seoul National University, Seoul, Korea
- B.S. in **Computer Science** — Oct 2012 - Aug 2017
  Sharif University of Technology, Tehran, Iran

Work Experience

- RBC Borealis, — Apr 2024 - Present
  Machine Learning Researcher,
  Contributing to research and development of machine learning models for real-world applications in finance.
- Adobe, — Jun 2023 - Nov 2023
  Research Scientist/Engineer Intern,
  Worked on a new method for automatically generating layout designs.
- Borealis AI (RBC Borealis), — Jan 2022 - May 2022
  Machine Learning Research Intern,
  Worked on Iterative Multi-scale Refining Transformers for Time Series Forecasting
- Faraadid Company, — Feb 2017 - Aug 2017
  Research and Development Engineer,
  Worked on finding online methods for object detection, considering the trade-off between speed and accuracy.

Publications and Preprints

- Ye Yuan, **Amin Shabani**, Siqi Liu, *"Embedding-based Context-aware Reranker,"* ArXiv Preprint, 2025.
- Ye Yuan, **Amin Shabani**, Siqi Liu, *"FACTS: Table Summarization via Offline Template Generation wtih Agentic Workflows,"* NeurIPS workshop on Generative AI in Finance (**NeurIPS Workshop**), 2025.
- **Amin Shabani**, Zhaowen Wang, Difan Liu, Nanxuan Zhao, Jimei Yang, Yasutaka Furukawa *"Visual Layout Composer: Image-Vector Dual Diffusion Model for Design Layout Generation,"* Conference on Computer Vision and Pattern Recognition (**CVPR**), 2024.
- Sepidehsadat Hosseini, **Amin Shabani**, Yasutaka Furukawa, *"PuzzleFusion: Unleashing the Power of Diffusion Models for Spatial Puzzle Solving,"* Thirty-seventh Conference on Neural Information Processing Systems (**NeurIPS**), 2023. (Spotlight)
- **Amin Shabani**, Sepidehsadat Hosseini, Yasutaka Furukawa, *"HouseDiffusion: Vector Floorplan Generation via a Diffusion Model with Discrete and Continuous Denoising,"* Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023.
- **Amin Shabani**, Amir Abdi, Lili Meng, and Tristan Sylvain, *"Scaleformer: Iterative Multi-scale Refining Transformers for Time Series Forecasting,"* Eleventh International Conference on Learning Representations **(ICLR)**, 2023.

- Weilian Song, Mahsa Maleki, **Amin Shabani**, Yasutaka Furukawa, *"Vectorizing Building Blueprints,"* Proc. 16th Asian Conference on Computer Vision **(ACCV)**, 2022.
- **Amin Shabani**, Weilian Song, Yasutaka Furukawa, Makoto Odamaki, Hirochika Fujiki, *"Extreme Structure from Motion for Indoor Panoramas without Visual Overlaps,"* Proc. International Conference on Computer Vision **(ICCV)**, 2021.
- **Mohammad Amin Shabani**, Laleh Samadfam, Mohammad Amin Sadeghi, *"Local Visual Microphones: Improved Sound Extraction from Silent Video,"* Proc. British Machine Vision Conference **(BMVC)**, 2017. (Spotlight)
- M. Abdolmaleki · J. P. Hutchinson · S. Gh. Ilchi · E. S. Mahmoodian · N. Matsumoto · **M. A. Shabani**, *"On uniquely k-list colorable planar graphs, graphs on surfaces, and regular graphs,"* Graphs and Combinatorics **(GCOM)**, Article No s00373-018-1879-7. (Authors are ordered alphabetically)

Patents

- Amin Shabani, Tristan Sylvain, Lili Meng, Amir Abdi, *"Multi-scale artificial neural network and a method for operating same for time series forecasting"*, US Patent App. 18/197,197.

Presented Seminars

- M. Abdolmaleki, S. Gh. Ilchi, E.S. Mahmoodian, and M. A. Shabani, *"On decomposing complete tripartite graphs into 5-cycles,"* **IPM**, Institute for Research in Fundamental Sciences, Iran, 2017. (Authors are ordered alphabetically)

Teaching Experience

Simon Fraser University

- Teaching Assistant for Introduction to Programming. — Summer 2020
- Teaching Assistant for Data Structures and Programming. — Summer 2020
- Teaching Assistant for Computer Vision. — Fall 2020
- Teaching Assistant for Communication Strategies. — Spring 2021
- Teaching Assistant for Data Structures and Programming. — Summer 2021
- Teaching Assistant for Introduction to Programming. — Fall 2021
- Teaching Assistant for Software Development Methods. — Fall 2022
- Teaching Assistant for Special Topics in Computing Systems. — Spring 2023
- Teaching Assistant for Special Topics in Network. — Spring 2023
- Teaching Assistant for Computer Vision. — Fall 2023

Sharif University Of Technology

- Teaching Assistant for Image Processing. — Fall 2016
- Teaching Assistant for Data Structures. — Fall 2014
- Teaching Assistant for Discrete Mathematics. — Spring 2014
- Teaching Assistant for Computer Programming. — Fall 2013

NODET High Schools

- Teaching Informatics Olympiad courses, NODET High Schools. — 2011 - 2017
- Teaching C++ Programming, NODET High Schools. — 2011 - 2017

Honors and Awards

- ◇ 1st Place Winner, Financial Agentic Retrieval Grand Challenge (FinAgentBench), AI for Finance Symposium '25 — Fall 2025
- ◇ Helmut & Hugo Eppich Family Grad Scholarship, SFU — Spring 2023
- ◇ SFU Computing Science Graduate Fellowship — Fall 2020 - 2022
- ◇ Helmut & Hugo Eppich Family Grad Scholarship, SFU — Spring 2021
- ◇ Graduate Scholarship for Excellent Foreign Students, SNU — Sep 2017 - Sep 2019
- ◇ SFU Graduate Fellowship — Fall 2019 - 2022
- ◇ SFU Research Assistant Scholarship — Spring 2020 - Spring 2023
- ◇ SFU Teaching Assistant Scholarship — Summer 2020 - Fall 2022
- ◇ Bronze Medal in the 20th Iranian National Olympiad in Informatics (INOI), Tehran, Iran. — Jul 2010

Extracurricular Activities

- ◇ Reviewer for ECCV2024, NeurIPS2023, SIGGRAPHA2023, ICCV2023, CVPR2023, ACCV2022, ICCV2021
- ◇ **Hamband**, Student council of department of mathematics, SUT. — 2014 - 2016
- ◇ **Algorithms and Problem Solving Laboratory**, under supervision of Prof. Yahya Tabesh, Sharif University of Technology, Tehran, Iran. — 2015 - 2016

Technical Skills

- ◇ *Languages & Frameworks:* Python, C++(STL), PyTorch, TensorFlow, Matlab